DGMS 2019 • SECTION K: GEOMORPHOLOGY & GLACIAL GEOLOGY

Question 81: Glacial Alpine Landforms: Arête vs. Col Differentiation

Official Examination Guide • Glacial Erosional Features, Terminology & Landform Identification

OFFICIAL EXAMINATION QUESTION

Which one of the following pairs is NOT correctly matched?

(a) Nunatak : Unglaciated island of bedrock, formerly surrounded by ice

(b) Cirque : Steep-walled semicircular basin in mountain glaciers

(c) Col : Narrow, sharp-edged ridge in mountain glaciers CORRECT ANSWER (MISMATCHED PAIR)

(d) Fjord : U-shaped glacial trough below sea level

1. Executive Summary & Core Principle

Alpine and continental glaciation produce distinctive erosional topography characterized by sharp crests, amphitheater basins, and drowned troughs. In classic geomorphology (Bloom, *Geomorphology*; Thornbury, *Principles of Geomorphology*; Mahapatra, *A Text Book of Geology*):

- **Arête (French for "fishbone" or "ridge"):** A sharp, steep-sided, knife-edged, serrated rock ridge that separates two adjacent glacial valleys or opposing cirques. It is formed when two adjacent valley glaciers actively scour and quarry the rock wall between them.

- **Col:** A mountain pass, gap, or saddle-like depression carved across an arête or mountain divide. A col forms when two cirques on opposite sides of a divide erode backward (headward sapping and plucking) until their headwalls intersect and breach the ridge crest. A col is thus a *pass/gap*, **NOT a narrow, sharp-edged ridge**!
- **Cirque (Corrie / Cwm):** A steep-sided, amphitheater-like, semicircular or horseshoe-shaped hollow excavated into high mountain slopes at the head of a glacial valley by freeze-thaw frost wedging, plucking, and rotational sliding of glacier ice. (Correctly matched in Option b).
- **Nunatak:** An isolated peak or island of exposed bedrock projecting above surrounding glacial ice or an ice sheet, unmodified by glacial ice cover. (Correctly matched in Option a).
- **Fjord (Fiord):** A deep, steep-walled, U-shaped glacial trough carved by valley glaciers below sea level, or submerged by post-glacial marine transgression. (Correctly matched in Option d).

Because Option **(c)** defines a *Col* using the morphological definition of an **Arête**, it represents the **incorrectly matched pair** and is the correct answer to the question.

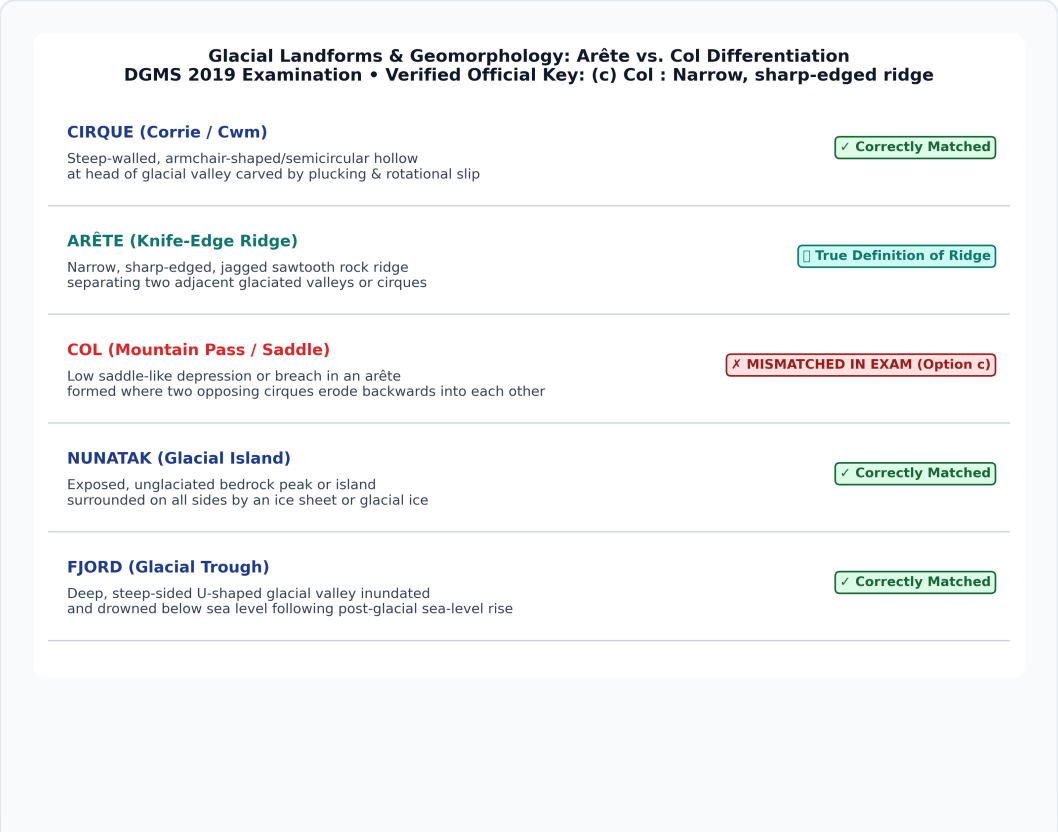


Figure 81.1: Schematic and morphological definitions of alpine glacial erosional landforms, contrasting the knife-edge crest of an Arête with the breached pass/ saddle of a Col (Grounded in Thornbury and Bloom Geomorphology).

2. Systematic Morphological Classification of Glacial Landforms

Landform	Etymology / Origin	Geomorphological Definition	Formation Process
Cirque (Corrie / Cwm)	Latin *circus* (circle); Scottish Gaelic *coire* (cauldron)	Armchair-shaped, steep-walled, flat-bottomed rock basin at glacier head.	Rotational ice slippage, plucking at headwall, nivation, and abrasion. Often holds a tarn lake.
Arête	French *arête* (fishbone, ridge)	**Narrow, sharp-edged, serrated, knife-like ridge.**	Parallel erosion by two adjacent valley glaciers or headward erosion of parallel cirques.
Col	Latin *collum* (neck); French *col* (pass)	**Saddle-shaped gap, breach, or pass along a ridge.**	Headward retreat and intersection of two opposing cirques, lowering the arête crest into a pass.
Horn (Pyramidal Peak)	German *Horn* (Matterhorn type)	Sharp, steep-sided, pyramidal peak with faceted faces.	Intersection and backward erosion of three or more cirques converging from different sides.
Nunatak	Inuit *nunataq* (lonely peak)		Glacial nunataks project above the ice surface,

Landform	Etymology / Origin	Geomorphological Definition	Formation Process
		Bedrock peak or ridge surrounded by glacier ice/ice sheet.	undergoing intense periglacial frost shattering.
Fjord (Fiord)	Old Norse *fjörðr* (passage)	Deep, drowned U-shaped glacial valley with steep rock walls.	Glacial valley excavated well below sea level (overdeepened by ice) subsequently flooded by post-glacial eustatic sea-level rise.

3. Evaluation of Options

- **Pair (a) [Nunatak] is CORRECTLY MATCHED:** An unglaciated rock island or peak surrounded by an ice cap/glacier.
- **Pair (b) [Cirque] is CORRECTLY MATCHED:** A steep-walled, semicircular armchair-shaped basin carved into mountain slopes.
- **Pair (c) [Col] is NOT CORRECTLY MATCHED (CORRECT ANSWER):** Defines a col as a "narrow, sharp-edged ridge". That is the definitive definition of an **arête**! A col is the saddle or pass formed by the breach of an arête.
- **Pair (d) [Fjord] is CORRECTLY MATCHED:** A submerged U-shaped glacial trough flooded by the sea.

> **Exam Summary:** Arête = Sharp ridge; Col = Saddle/pass breach in the ridge; Horn = Pyramidal peak; Cirque = Semicircular basin; Nunatak = Ice-surrounded rock peak. Therefore, **(c)** is the mismatched pair.

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